# Tier 1 Lab: RC and RLC Circuits

Capacitors and capacitance, RC circuits, time constants, RLC circuits.

## Objectives

- Build circuits and measure the characteristics of those circuits using an oscilloscope.
- Construct a capacitor and measure capacitance.
- Apply error analysis skills to measurements.

## Overview

In the EM2 Lab, you analyzed AC circuits. In this lab, we will use the oscilloscope to study the time-dependent behavior of both RC and RLC circuits with a DC source and build a capacitor. You will first explore charging and discharging behaviors of RC series circuits. Then, you will explore the transient behavior of RLC circuits. Finally, you will construct your own capacitor with household materials and measure its capacitance.

## What to Turn In

The **group as a whole** will submit a single copy of the lab report pdf, with all measurements, calculations, error propagation derivations, graphs and question answers included. Lab report specifics for the Introduction/Objectives, Theory, Methods, Calculation and Data Reduction-Analysis, Summary and Conclusion sections are listed in the Lab Report document**.** Each group member's contribution must be clearly stated throughout the lab report. A separate bCourses submission is required for every Python or Excel file used in the analysis**.**

All analysis work **must be equally** distributed in the group and properly attributed. The group, as a whole, decides which datasets each person will analyze.

# Theory and Background

## Capacitors, Capacitance, and Breakdown Voltage

In the previous lab we defined a ***capacitor*** as a two-terminal device consisting of two conductors separated by a dielectric layer that stores electrical charge and electrical energy. When the conductors carry a charge difference of $Q$ there is a voltage difference $V_C$ between the conductors. The ***capacitance*** of a capacitor is defined as the ratio of charge-to-voltage,

$$C = \frac{Q}{V_C},$$

The simplest kind of capacitor is the ***parallel-plate capacitor*** consisting of two plates of equal area $A$ held parallel to each other, separated by a width $d$ of dielectric material with relative static permittivity (or dielectric constant) $\varepsilon_r$. In this case, the capacitance is

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}.$$

If the electric field strength inside a dielectric exceeds a break-down threshold strength $E_{bd}$ then ***dielectric break-down*** occurs, which means that charges suddenly start flowing through the dielectric.[1] Real-world capacitors[2] are rated by their ***break-down voltage*** (or ***threshold voltage***) $V_{bd} = E_{bd}\, d$. Applying a voltage greater than $V_{bd}$ may cause break-down and may permanently damage the capacitor and circuit.

## RC Circuits with DC Source

The RC Circuit in this lab consists of a capacitor $C$ and a resistor $R$ connected in series to a DC source $V_0$, as shown.

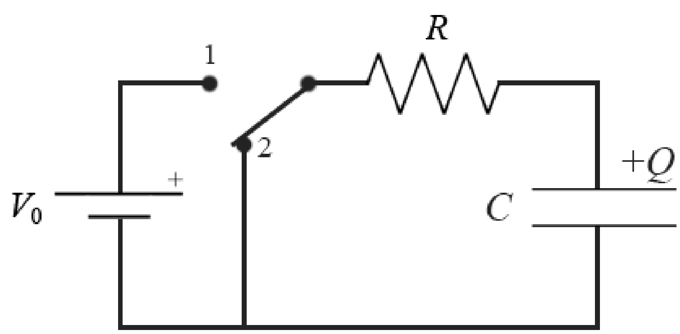


*Figure 1 Charging and discharging a capacitor in an RC circuit.*

Initially, the switch has been in Position 2 for a long time so that the capacitor is completely discharged ($Q$ = 0). When the switch is toggled to Position 1, current starts to flows out of the battery and begins to charge the capacitor. As charge accumulates on the capacitor, the potential across it increases. The current decreases until the potential across the capacitor equals the battery emf. When the switch toggles back to Position 2, the capacitor begins discharging through the resistor.

### Discharging Capacitor (Switch Position 2)

For the discharging capacitor, Kirchhoff's voltage law for the loop gives

$$-V_R - V_C = 0.$$

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[1] Examples of dielectric breakdown of air include sparks and, more dramatically, lightning.
[2] https://en.wikipedia.org/wiki/Capacitor#Non-ideal_behavior

Using Ohm's Law ($V_R = IR$), the relationship between voltage and charge in a capacitor ($V_C = Q/C$), and the relationship between charge and current ($I = dQ/dt$) gives a linear, first-order ordinary differential equation for $Q$,

$$\frac{dQ}{dt} = -\frac{1}{RC}Q.$$

For the discharging capacitor we start with an initial condition $Q(0) = Q_0 = CV_0$ which gives the solution

$$Q(t) = Q_0 e^{-t/RC} = CV_0 e^{-t/\tau},$$

where we have defined the ***time-constant***

$$\tau \equiv RC \tag{1}$$

The capacitor voltage is therefore

$$V_C(t) = V_0 e^{-t/\tau} \tag{2}$$

Note that this circuit reaches an asymptotic steady-state with $\lim_{t\to\infty} Q(t) = 0$ and $\lim_{t\to\infty} I(t) = 0$.

## Charging Capacitor (Switch Position 1)

Following a similar process as above, when the switch is in position 1 we have a charging-capacitor setup. The Kirchhoff's voltage law for the loop gives

$$V_0 - V_R - V_C = 0$$

This results in another linear, first-order ordinary differential equation for $Q$,differential equation

$$\frac{dQ}{dt} = -\frac{1}{RC}Q + \frac{V_0}{R}.$$

For the charging capacitor we start with initial condition $Q(0) = 0$ which gives the solution

$$Q(t) = CV_0\left(1 - e^{-\frac{t}{\tau}}\right),$$

where $\tau$ is the same time constant defined in Eq. 1. The capacitor voltage is therefore

$$V_C(t) = V_0\left(1 - e^{-\frac{t}{\tau}}\right). \tag{3}$$

Note that this circuit reaches an asymptotic steady-state with $\lim_{t\to\infty} Q(t) = CV_0$ and $\lim_{t\to\infty} I(t) = 0$. A graph of voltage-vs-time for a charging capacitor followed by a discharging capacitor is shown in Figure 2.

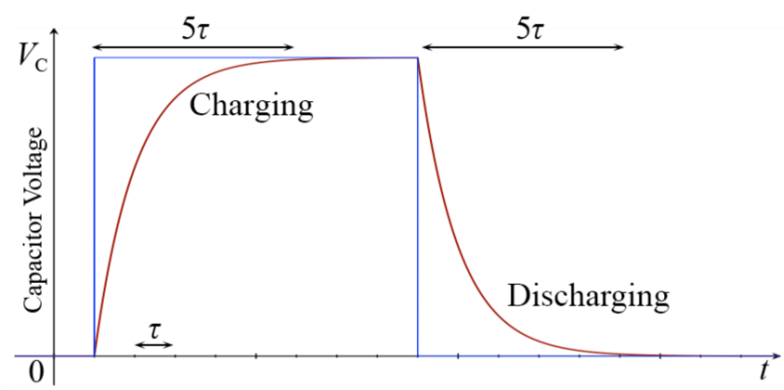


*Figure 2. Charging and discharging behavior of a capacitor. Blue shows the applied DC voltage and red shows the capacitor voltage.*

## Series RLC Circuits

If an inductor, a capacitor and a resistor are connected in one circuit, energy will transform between the electric field energy of the capacitor and the magnetic field energy of the inductor, while being dissipated by the resistor. As a result, such RLC circuits will behave as the electronic counterpart of a damped harmonic oscillator.

***Transient behavior*** is observed when the circuit input has a short-lived amplitude change and is then held constant, which occurs when a switch connected to a DC source is closed/opened. So, let us consider a series RLC circuit with a capacitor, a resistor, an inductor, and a DC voltage source are connected in series as shown in Figure 3. The voltage across the capacitor, inductor and resistor are $V_c$, $V_I$ and $V_R$, respectively, and the source voltage is $V_0$. The charge on the capacitor is $Q$.

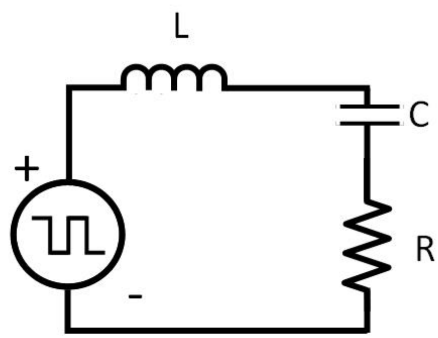


*Figure 3. A series RLC circuit connected to a DC source.*

Kirchhoff's voltage law for the loop, when a DC voltage $V_0$ is applied is:

$$V_0 - V_R - V_L - V_C = 0.$$

The voltages for the circuit components are given by $V_R = IR,$ $V_L = L\, dI/dt,$ and $V_C = Q/C.$ Combining these gives a linear, inhomogeneous, second-order differential equation for $Q$,

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = V_0.$$

We will be concerned primarily with the voltage across the resistor and thus the *current I* flowing in the circuit. Taking the derivative of this equation with respect to time gives a linear, homogeneous, second-order differential equation for $I$,

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{I}{C} = 0. \tag{4}$$

If you recall our previous labs, this equation looks exactly like the equation describing the damped mechanical harmonic oscillators

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0,$$

where current plays the role of displacement, inductance of the mass, resistance of the friction, and capacitance of the inverse spring constant. In analogy to that system we define two parameters with dimensions of inverse time that will determine the behavior of the system,

$$\omega_0 \equiv \frac{1}{\sqrt{LC}}, \qquad \alpha \equiv \frac{R}{2L}. \tag{5}$$

The solutions to equation (4) fall into three different regimes of behavior based on the ratio $\alpha/\omega_0$ - ***underdamped*** when $\alpha < \omega_0$, ***critically damped*** when $\alpha = \omega_0$, and ***overdamped*** when $\alpha > \omega_0$. The underdamped case will exhibit oscillations with angular frequency $\omega = 2\pi f$ and the overdamped case will exhibit exponential decay with decay parameter $b$,

$$\omega = \sqrt{\omega_0^2 - \alpha^2}, \qquad b = \sqrt{\alpha^2 - \omega_0^2}.$$

The behavior of the current in each of the three regimes is summarized below [*see Purcell, Chapter 8, Section 1 for a detailed derivation*].

Underdamped ($\alpha < \omega_0$) $$I(t) = e^{-\alpha t}(A\cos\omega t + B\sin\omega t) = \tilde{A}\, e^{-\alpha t}\cos(\omega t - \varphi), \qquad (6a)$$

Critically damped ($\alpha = \omega_0$) $$I(t) = e^{-\alpha t}(A + Bt), \qquad (6b)$$

Overdamped ($\alpha > \omega_0$) $$I(t) = e^{-\alpha t}(Ae^{+bt} + Be^{-bt}). \qquad (6c)$$

In other words, if the RLC circuit is connected to a constant voltage source, energy will oscillate between the capacitor (stored in the electric field) and inductor (magnetic field), while being dissipated (damped) by the resistor. As already familiar to us from mechanics, the ratio between $\alpha$ and $\omega$ determines the damping regime of the damped harmonic oscillator (see Figure 4):

- "Underdamped" when $\alpha < \omega_0$. The curve oscillates as a sine wave at frequency $\omega_0$, with the peaks decaying with an exponentially as $\alpha$. In Eq. 6a we write the solution in two different forms. In the first form the two initial conditions determine the amplitudes $A$ and $B$ of the cosine and sine wave components. In the second form the two initial conditions determine the overall oscillation amplitude $\tilde{A}$ and the phase angle $\varphi$.
- "Overdamped" when $\alpha > \omega_0$. The aperiodic exponential waveform decays with no oscillations.
- "Critically damped" when $\alpha = \omega_0$. The non-oscillatory decay is given by $I(t) = e^{-\alpha t}(A + Bt)$ and goes to zero faster than any overdamping case.

In every regime, the constants $A$ and $B$ are determined from the initial conditions.

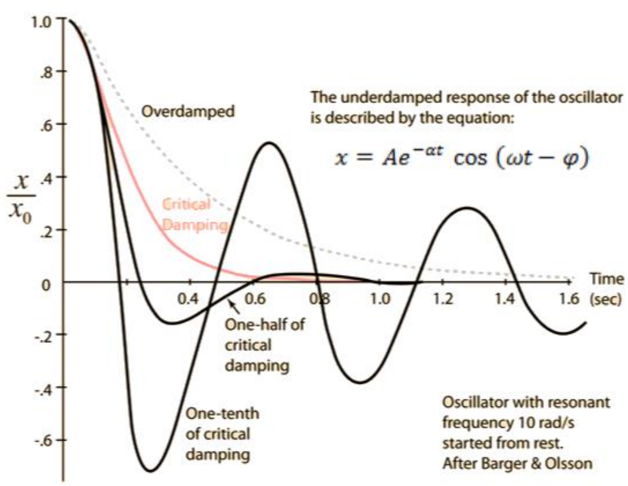


*Figure 4. Overdamping, critical damping and underdamping with the same intial conditions. Adapted from [http://hyperphysics.phy-astr.gsu.edu/](http://hyperphysics.phy-astr.gsu.edu/).*

# Equipment and Equipment Notes

- MSO 2024B Oscilloscope
- Tektronix AFG2021 Function Generator
- Fluke 179 Digital Multimeter
- Breadboard
- Wires, Cables, and Adapters
  - Test leads, mini-grabber
- Circuit Elements
  - Resistors: 100 Ω, 1 kΩ, 10 kΩ, 100 kΩ ± 1%
  - Capacitor: 0.1 μF, **47 μF ± 5%**
  - Inductor: 100 mH ± 10%
- Material for Homemade Capacitor
  - Aluminum foil, glue stick, paper, etc.

Recall that the inductor has an internal resistance $R_L$ that you will need to measure.

# Part 1: Measuring the Time Constant Using the Oscilloscope

We will use a function generator to obtain a square wave – fixed periods of constant voltage followed by fixed periods of zero voltage. This allows you to replace the battery in Figure 1 with a square wave. We will be using the oscilloscope to make detailed measurements of the waveform.

***In order to effectively capture the exponential behavior, we will want each segment of the square wave to last at minimum 5τ (five time constants).*** [3]

The graphs in Figure 2 were created using a time of $8\tau$ for each phase of the square wave (a total period of $16\tau$). It is often good to do a quick estimation of the time constant $\tau$ to make sure everything looks alright.

*Hint: To estimate the time constant from your graph, do a quick measurement of two points in the curve, and plug them into the formula 2 or 3..*

## Equipment Setup

### Function Generator

First we will set the function generator to create a low-frequency square wave to serve as our power source.

- Choose "Square Wave" as your desired signal output.
- Set the frequency to 100 mHz using the menu.
- Set the output load impedance to Hi-Z. In "Amplitude/Level" menu
- Set the "Amplitude" to 10 Vpp.
- Set the "Offset" to 5V. (*Think: why we are setting the offset to half the amplitude?*)
- Connect a BNC T-splitter to the leftmost function generator output.
- You now are ready to feed this signal to your oscilloscope and your RC circuit. Turn the output "On"

[3] A question you should consider: Why $5\tau$? Why not $0.5\tau$, $1\tau$, or $20\tau$?


*Figure 5. Signal generator setup*

## Oscilloscope

- Display channels 1 and 2 on the scope.
- Connect one BNC cable from the T-splitter to the oscilloscope CH1 input.
- Connect a mini-grabber to the remaining BNC cable from the other end of the T-splitter.
- Connect a BNC cable from the oscilloscope CH2 input to a different mini-grabber.

## RC Circuit

- Set up the RC circuit shown in Figure 6 on the breadboard, using an $R = 10\ \text{k}\Omega$ resistor, a $C = 47\ \mu\text{F}$ capacitor, and the signal input from the T-splitter BNC/mini-grabber cable.
- Connect the mini-grabber cable from CH2 across the capacitor.

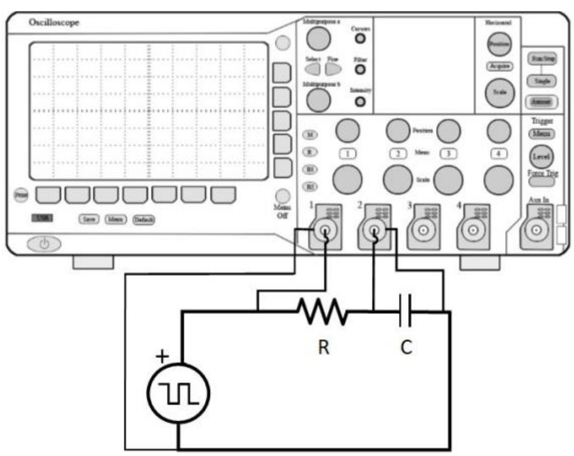


*Figure 6. RC circuit connected to an oscilloscope.*

## Establish the oscilloscope triggering

The oscilloscope cannot trigger repetitively in Auto mode because the signal is so slow. Instead, it will be operating in “roll mode”, functioning like an analog strip chart. The time base, rather than moving in response to triggering as in conventional mode, remains static and the waveform moves. Accordingly, in roll mode, the trace enters at the right of the of the display and travels slowly across the screen at a speed determined by the frequency of the signal and the horizontal scale in units of time per division.

- Adjust the CH1 and CH2 amplitudes to 2V and the time scale knob to 1.0s. You should observe CH1 and CH2 signals rolling across the screen.

  *If a rolling signal is not observed, the scope settings are incorrect. Try setting the signal generator frequency to 20 Hz, use Auto to trigger, then slowly decrease the frequency to 100 mHz*

- Press the "Run/Stop" button to freeze the display on your screen. The trace is now stored in scope memory until the Run/Stop button is pressed again. It should generally resemble Figure 7.

- Sometimes you need to examine the stored signal to see its details. The oscilloscope controls have the same function on the stored signal as on a triggered signal.  As an example, to zoom on the discharge part of the signal, use the time scale knob to stretch the signal and the horizontal position knob to move on the screen.

  *Note: Set the CH1 and CH2 amplitudes to fill the oscilloscope display (do not overfill) at whichever scale you choose*

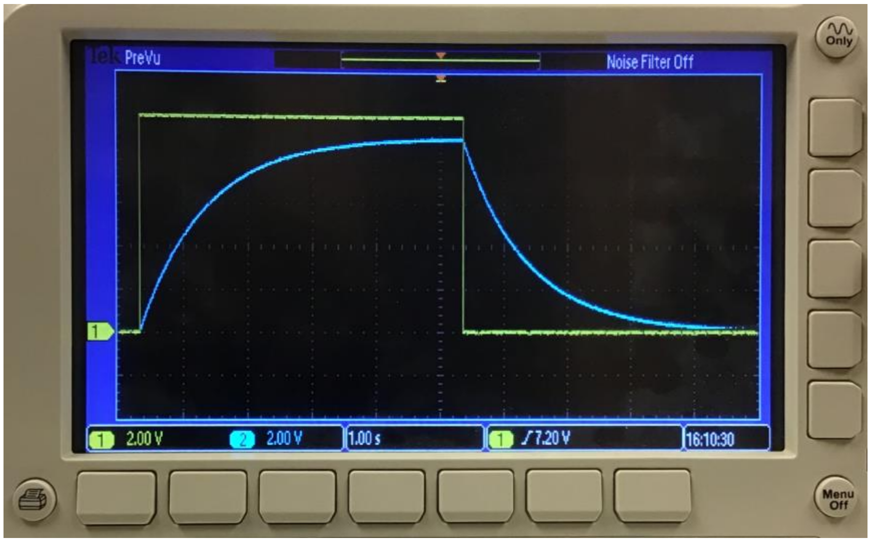


*Figure 7. Output signal of the signal generator (yellow color) is displayed on channel 1, and the voltage across an RC circuit capacitor (blue color) is displayed on channel 2.*

## Charging and discharging decay measurement procedure:

- With the oscilloscope in roll mode, set the horizontal cursors to the maximum and minimum voltage levels. Please note that your capacitor may not charge to the maximum supply voltage of 10V and the capacitor may not discharge to zero.  ***Record the voltage values***.

- Set the time base to 400 ms and using the Run/Stop button, capture a decay (or charge) curve. You should see a signal close to what is displayed in Figure 8 except the location of your cursors may be different.

- You will be using the cursors for all measurements.  Review the cursor tutorial from the previous lab.

- Place one of the horizontal cursor lines (using either Multipurpose A or B) at the level where the discharge (or charge) starts and set its vertical line to the beginning of the signal change, where the signal starts to decay (or rise). This will be your reference point, and you are going to measure voltage drops and decay time (or voltage gain and growth time) using this reference point.

- ***Record the reference cursor values in case you accidentally move the cursor.***

- Set the second cursor horizontal line to the desired voltage change, then the vertical line to intersect the decay (charging) curve.

- Record the location of the changing cursor using the information box on the top right of the display screen on the scope, both the actual values and the
- You might need to change amplitude/time scales to better measure the decay curve. If you redefine your reference cursor location, record in your lab notebook.
- It is also possible to export the data in a raw format (>100K points) directly from the oscilloscope to a USB thumb drive. Whichever technique you use clearly label your data.


*Figure 8. Measuring the discharge time across the capacitor after the initial value has dropped 2V using the cursors.*

## Experiment 1A: Measuring the RC Time Constant ($R$ = 10 kΩ, $C$ = 47 μF)

**Goal:** Measure the RC time constant using the discharge and/or charge curves for the specified components, using the circuit in Figure 6, for performing a model fit.

**Setup:** Arrange the resistor in series with your capacitor, as shown in Figure 6.

**Measure:** The group should choose how to characterize the discharging (or charging) behavior. Develop a data collection protocol using the “Charging and discharging decay measurement” procedure. Acquire enough points to perform a model fit to the measured curves, approximately 10 data points.

**Hints and Reminders:**

- *Record the maximum and minimum voltages for both channels.*
- *Establish your CH2 start time/voltage reference using the CH1 voltage transition for the capacitor discharging (or charging),*
- *At the beginning, record the voltage drop more carefully (~0.5V), then less frequently at later times. (Why?) Continue finding times/voltage until you reach the minimum value.*

**Analysis:** [In class] Do a quick estimation of the time constant $\tau$ to make sure everything looks alright.

*Hint: What do you expect the time constant to be with this circuit?*

## Data Analysis for Experiment 1A

- Use the data from Experiment 1A to determine the discharge time constant. You may either linearize the data or do a direct non-linear fit, whichever method works better for you.

- **Determine the experimental values of the amplitude *A* and time constant *B*** using the generalized fit equations $V_C(t)/V_0 = Ae^{-t/B}$ (for discharging) *or* $V_C(t)/V_0 = A\left(1 - e^{-\frac{t}{B}}\right)$ (for charging), where $V_0$ is the 10V supply voltage, to solve for the fit amplitude $A$ and the fit time constant $B$, and the associated errors.

  *Hint: Be sure to define good data time range and voltage zeros.*

- **Compare the fitted *B*** (each group member calculates using their own data) with the theoretical time constant $\tau$ for each circuit using an agreement test.

  *Hint: Use τ = RC, where R is the nominal component resistance with error, C is the component capacitance with error measured with the Fluke, and the error is properly propagated. If you do not find agreement, what are the possible reasons?*

➢ Be sure to include plots of your data and the fits, with appropriate labeling!

## Experiment 1B: Handcrafting a Capacitor

Now it is time to build your own capacitor! As mentioned previously, a capacitor consists of essentially two conductors separated by an insulating layer. Therefore, if you glue two sheets of aluminum foil to both sides of a piece of paper and wire it, this Aluminum-paper-Aluminum sandwich can be a reasonable capacitor!

### Estimation

First, let's do some back-of-envelope calculations. It is always good practice to estimate something's viability to some extent before building it. Assuming you are using a standard US letter size paper (215.9 mm x 279.4 mm) with thickness 0.1 mm, ignoring the glue thickness, how much capacitance will your capacitor have? Recall that $C = \varepsilon_r \varepsilon_0 A/d$ , where $\varepsilon_r = 3.8$ for paper, $\varepsilon_0 = 8.854 \times 10^{-12}$ F ·m$^{-1}$ is the vacuum permittivity, $A$ is the overlapping area of the plates, and $d$ is the separation between the plates.

### Handcrafting

What follows is just one possible way to do this. The essence of DIY is to come up with your own way (though please be safe)!

***For this part every student needs to construct their own capacitor, though you will still work in groups.***

*Figure 8. An example of a homemade capacitor.*

**Goal:** Make a capacitor with the capacitance as large as possible. [As you will find out, its capacitance tends to be smaller than your estimation.]

- Plan out the capacitor prior to beginning. How will you make the electrical connections? What is a good construction method? What should the final capacitor be placed during the measurements?

- Cut two pieces of aluminum foil and glue them to both sides of a piece of printer paper. Since capacitance is inversely proportional to separation distance between the two plates, it is very important to make sure the foils are flat and stick very well to the paper. Also make sure there is no pinhole in the paper since it will short your capacitor.

- Electrically connect to the two aluminum foils respectively. You can tape a jumper wire or use the alligator clip for the connection.
- Measure the resistance between the jumper wire and foil to make sure they are in good contact (**less than 1Ω**), then check the other side.
- Check the capacitance using your DMM.
- Do a back of the envelope calculation to find an order of magnitude estimate for the capacitance possible with your size of capacitor and materials. Do any of your measurements contradict this estimate? Sometimes the wet glue used to create the capacitor causes the DMM to measure unreasonably high capacitances. Be sure to discuss this in your report.

You may find it is difficult to fully charge your capacitor. This may come as a result of losing charges to its contacting surface.

- Try rolling it up (adding paper on either side might be useful) or anything else you can come up with, to deal with this.
- It is OK to construct several capacitors, until one works.

Capacitor construction references include:

  - The Physics Teacher 52, 482 (2014); https://doi.org/10.1119/1.4897586
  - instructables.com/aluminum-foil-plate-capacitor/

### Determining the Capacitance

**Setup:** Construct the RC circuit in Figure 6 using your homemade capacitor and a 10 kΩ or 1 MΩ resistor.

**Measure:** Take data that will allow you to measure the capacitance of your home-built capacitor using the charging and discharging curves with an oscilloscope. You may have to try several times before you manage to measure it. *Take a picture*!

*Hints: You may need to use a different value of resistance to get a good measurement.*

*Five or fewer curve points are enough to obtain $\tau$ and characterize the behavior*

## Data Analysis for Experiment 1B

**Goal:** Analyze the charging and discharging behavior of your group's home-made capacitor.

- **Derive the capacitance** of your home-made capacitor by performing a fit using your data.
- How is it **different** from your estimation, and what are the possible reasons?
- **Estimate an effective dielectric thickness** of your home-made capacitor.
- As you have probably found out, it is difficult to achieve a large capacitance! What is the industrial trick to pack 1 mF or even 1 F capacitance into such a small commercial capacitor? Figure out the answer online, and write down a summary in the lab report.

# Part 2: RLC Circuit Transient Behavior

## Setup

Experiment 1 explored the quasi-DC time dependence of two circuit elements, a resistor and capacitor. When an inductor is added, new phenomena of the resonant circuit can be observed. In Experiment 2, we explore the transient behaviors of RLC circuits in the electronic harmonic oscillator under and over damped regimes by changing the circuit components. First, the underdamped circuit will be explored. Each group then can explore the overdamped regime through appropriate circuit design.

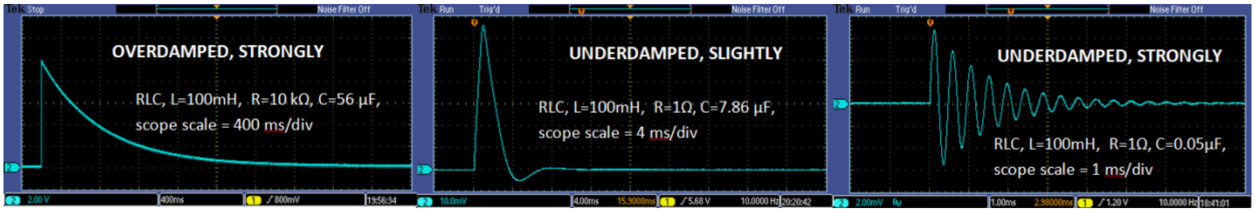


*Figure 9. AC RLC circuits measured with an oscilloscope, components as labeled.*

The over and underdamped cases will be studied in this Experiment. The middle case is included for illustration.

**Estimation:** The relationship between the parameters $\omega_0$ and $\alpha$ determines the operational characteristics for each resonant RLC circuit. When $\alpha < \omega_0$ the circuit is in the underdamped regime, when $\alpha = \omega_0$ it is critically damped, and when $\alpha > \omega_0$ it is in the overdamped regime. You will need to decide which regime (underdamped, critically damped, or overdamped) each circuit is in, using the definitions of $\omega_0$ and $\alpha$ from Eq. 5. In this $R$ is the *total* circuit resistance, equaling the sum of the circuit component resistor(s) and the inductor internal resistance $R_L$.

**Function Generator:** Hi-Z output impedance, 10 Hz square wave, 4 $V_{pp.}$ (you might need to know that the output impedance is 50 Ω regardless of the setting of the function generator, i.e. the Hi-Z setting only affects how the user-input input is translated to the output).

**Circuit:** Measure the inductor resistance before setting up the circuit. Always include it in the total resistance in the rest of the lab.

**Oscilloscope:** Auto trigger, CH1 records signal generator output, CH2 records the voltage across the resistors.

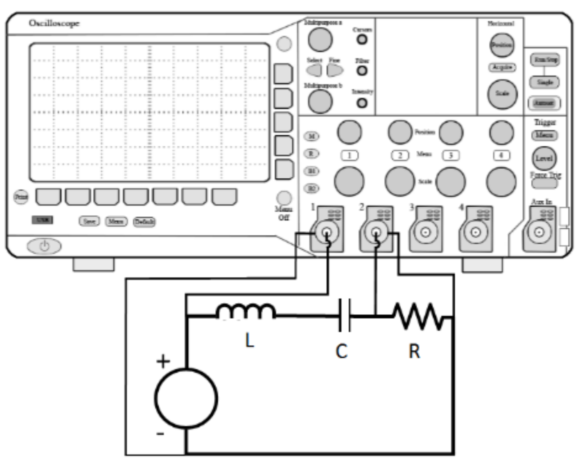


*Figure 10. The transient RLC circuit setup.*

## Experiment 2A: RLC Underdamped Transient Response

**Goal:** Explore the RLC underdamped transient response.

**Setup:** Connect the RLC circuit as shown in Figure 10 using a 100 mH inductor, 0.1 µF capacitor, and a 50 Ω resistor (two 100 Ω resistors in parallel). Record your experimental protocol and include a detailed circuit diagram and/or a picture of the circuit.

**Oscilloscope:** CH1 records the function generator output, CH2 records the voltage across the resistors. You should see the damped "ringing" behavior in the CH2 resistor voltage at both the rising and falling edge of the CH1 input square wave.

**Measure:** Record the observed waveform or take a picture. Record amplitude values of the CH2 peaks and their time relative to the beginning of either the rising or falling edge of the square wave input pulse, and label them on the sketch.

*Note: The data also can be downloaded to a USB flash drive for further analysis.*

## Data Analysis for Experiment 2A

**Goal:** Analyze the RLC underdamped transient response.

- **Determine the experimental values of $\alpha$ and $\omega_0$** by performing a fit using either form of the underdamped current behavior from Eq. 6a.
  - If you are recording the peak amplitudes manually it is probably best to use the second form (amplitude and phase angle).
  - If you are using your downloaded data so you have a sufficient number of points for a full fitting, you can use the first form.
- **Compare the experimental value of $\alpha$ and $\omega_0$ with the theoretical values**, using agreement tests.
- **If you do not find agreement**, what are the possible reasons?

## Experiment 2B: RLC Overdamped Transient Response

**Goal:** Explore the RLC overdamped transient response.

**Setup:** Connect the RLC circuit as shown in Figure 10 using a 100 mH inductor, 47 µF capacitor, and a 1 kΩ resistor. Record your experimental protocol and include a detailed circuit diagram and/or a picture of the circuit. Set the function generator frequency to a 50 or 100 mHz square wave.

**Measure:** Record the observed waveform or take a picture. Record "how fast" it dies down (it is up to you to determine what this means). Obtain enough points to characterize the decay constant.

*Note: The data also can be downloaded to a USB flash drive for further analysis.*

**Analysis:** [In class and in lab report] Does your recording resemble the Figure 9 left-most curve? Discuss.

## Data Analysis for Experiment 2B

**Goal:** Analyze the RLC overdamped transient response.

- **Determine the experimental decay time constant**. You may either linearize the data or do a direct non-linear fit, whichever method works better for you. Use the generalized fit equation $V_R(t)/V_0 = Ae^{-Dt}$, where $V_0$ is the supply voltage, to solve for the fit amplitude $A$ and the fit decay constant $D$, with the associated errors. Be careful - the $t = 0\ sec$ start time can be difficult to ascertain!
- **Compare the fitted *D*** with the theoretical decay constant(s) $\alpha \pm b$ for the circuit using an agreement test.
- **If you do not find agreement**, what are the possible reasons?